Convolutional networks

Flowers102

LECTURE 12 | GÉRON CH. 12

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

Where we are

Three lectures of neural networks, on flattened 28×28 greyscale images:

- Lecture 9 — a layer is $\phi(\mathbf{X}\mathbf{W} + \mathbf{b})$, and without ϕ a stack collapses
- Lecture 10 — you own the training loop, and reverse-mode differentiation is what makes it affordable
- Lecture 11 — a deep stack trains only if the variance survives the descent

WHAT BREAKS WHEN THE IMAGE GETS BIGGER

Flattening a 224×224 colour image gives 150,528 inputs. A single dense layer of 1,000 units on that is **150 million parameters** — for one layer, on one image size. Every one of them has to be learned from data, and none of them knows that two adjacent pixels are related.

Today's method fixes both problems at once, and the derivation says exactly how much it saves.

Today

1. The task, and why a flattened image is the wrong representation (10 min)
2. **The mathematics** — weight sharing, equivariance, and where the memory goes (20 min)
3. **The method** — convolution, pooling, an architecture, and training it (40 min)
4. What it costs, and what it cannot do (15 min)

PART ONE

The brief

15 minutes · the problem, the data, the stakeholder

The stakeholder

A horticultural wholesaler receives mixed pallets of cut stems and has to sort them by species before they reach the auction floor.

- One overhead camera per lane, one photograph per stem
- A human inspector currently does this and is the bottleneck
- They want the machine to name the species from the photograph

Same shape of problem as the garment sorter in Lecture 9. Everything that makes it harder is in the pixels.

What they actually asked for

> **IN THEIR WORDS**
>
> “Tell us which of our species it is. If you cannot tell, say so and we will look at it ourselves — but do not tell us the wrong one.”

Two requirements hiding in one sentence: a classifier over a fixed list of species, and a usable notion of *confidence*.

We build the first today. The second waits until the numbers are good enough to be worth calibrating.

Why the previous recipe will not survive contact

	Fashion MNIST	Flowers102
Image	28 × 28, grey	variable, colour
Classes	10	102
Training images	60,000	✗ 1,020
Images per class	6,000	✗ 10
Subject fills the frame	always	sometimes

Ten times the classes, 59 times fewer labelled images.

The dataset

`torchvision.datasets.Flowers102` — 8,189 photographs of 102 flowering species common in the United Kingdom.

```
from torchvision.datasets import Flowers102

train = Flowers102("data", split="train", download=True)
val   = Flowers102("data", split="val",   download=True)
test  = Flowers102("data", split="test",  download=True)
```

About 345 MB. It is one of the datasets the textbook names for this chapter, and the split below is the one shipped with it — so every result in the literature is on the same 6,149 test images.

What the camera sees

12 of the 102 species, one image each, resized to 128 × 128

Different scales, different backgrounds, different lighting — and several of these species are hard to tell apart by eye.

What is hard here, precisely

- **Position** — the flower is not centred, and not at a fixed scale
- **Background** — leaves, soil, other flowers, a gardener's hand
- **Between-class similarity** — several of the 102 species are separated by petal count and little else
- **Within-class variation** — the same species at two stages of opening

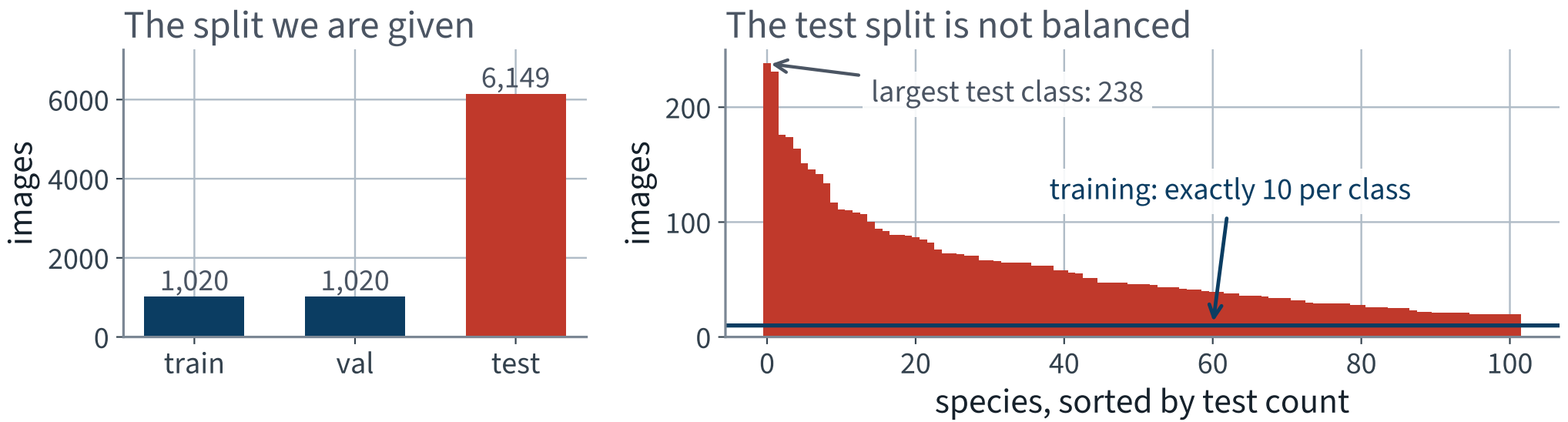
Only the first of those has a structural answer. It is the answer this chapter is about.

The split we are given

Split	Images	Per class	Used for
train	1,020	10, exactly	fitting
val	1,020	10, exactly	choosing
✗ test	✗ 6,149	20–238	reporting, once

✗ There are six times as many test images as training images.

Balanced where it is measured, not where it is reported



The training split is balanced by construction. The test split is not, and that will matter when we choose a baseline.

Why the training split is so small

Because somebody had to look at every photograph and name the species.

> **THE STANDING CONSTRAINT OF APPLIED VISION**
>
> Images are cheap and labels are not. Every application in this part of the course has more pixels than it can use and fewer labels than it wants.

Design for that. A method that needs thousands of labelled examples per class is not a method you can deploy here.

Two decisions before any model

1. **A fixed size.** Every image is resized to 128 × 128. A network with a dense layer needs a fixed input length, and 128 is the largest square this build fits into a free Colab session
2. **A fixed scale.** Pixels are divided by 255 and then standardised per colour channel

```
tf = transforms.Compose([
    transforms.Resize((128, 128)),
    transforms.PILToTensor(),        # uint8, (3, 128, 128)
])
```

The normalisation statistics

```
x = X_train.float() / 255.0          # the TRAINING split only
mean = x.mean(dim=(0, 2, 3))         # one number per colour channel
std  = x.std(dim=(0, 2, 3))

def normalise(x_u8):
    return (x_u8.float() / 255.0 - mean[:, None, None]) / std[:, None, None]
```

Channel	mean	std
red	0.4330	0.2896
green	0.3819	0.2408
blue	0.2964	0.2684

The obvious thing to try first

You have a multilayer perceptron that works. Flatten the image and feed it in.

	Count
Inputs, 128 × 128 × 3	49,152
Outputs, 32 values at every pixel position	524,288
✗ Weights in one fully connected layer between them	**✗ 25,769,803,776**

✗ That is 96 GB of float32 for one layer. The full argument is this lecture's mathematics block; here it is just a reason to do something else.

THE MATHEMATICS

Weight sharing, equivariance, and where the memory goes

20 minutes · examinable

The question

You replaced a matrix multiplication with a convolution and the network became trainable.

What exactly did you buy, and what did you give up?

> **THREE ANSWERS, IN ORDER**
>
> A parameter count. A symmetry. And a memory bill that is not where anyone expects it to be.

1 · The parameter count, on the same image

Fix the input and the output shape so the comparison is between two ways of computing *the same sized thing*.

	Shape	Values
Input	3 × 128 × 128	49,152
Output	32 × 128 × 128	524,288

The first layer of the network you built in the previous lecture, and the fully connected layer that would produce the identical output shape.

The dense layer

Every output value is connected to every input value:

$$\#\text{weights} \;=\; n_{\text{in}} \times n_{\text{out}} \;=\; (3HW)(32HW) \;=\; 96\,H^2W^2$$

✗ 25,769,803,776 weights.

At four bytes each that is 96 GB, for one layer, before any gradient or optimiser state. No accelerator in the building holds it.

The convolutional layer

Every output value is connected to a $k \times k$ window of every input channel, and **the same weights are used at every position**:

$$\#\text{weights} \;=\; C_{\text{out}} \times C_{\text{in}} \times k \times k \;=\; 32 \times 3 \times 7 \times 7$$

✓ 4,704 weights.

H and W do not appear. That is not an approximation or a simplification — the image size is genuinely absent from the count.

The ratio

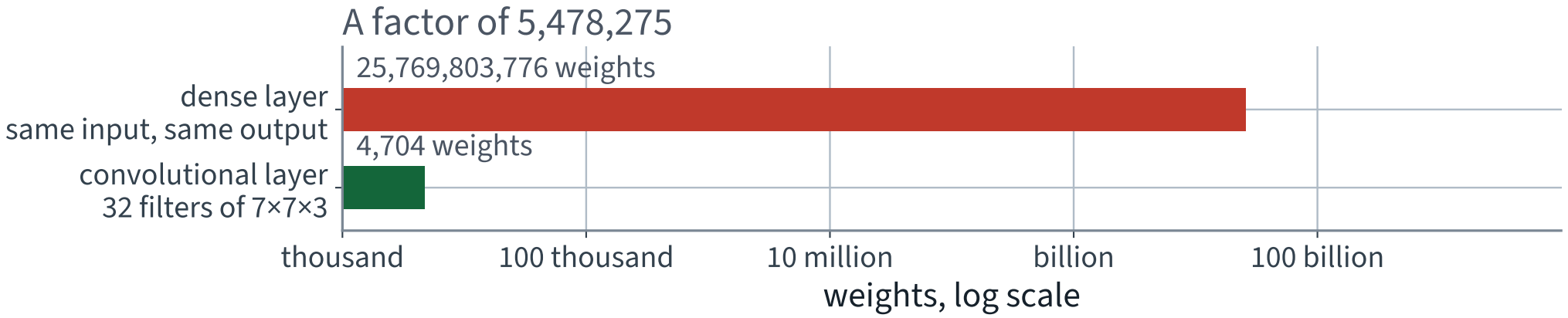
$$\frac{n_{\text{in}} n_{\text{out}}}{C_{\text{out}} C_{\text{in}} k^2} = \frac{H^2 W^2}{k^2}$$

5,478,275

times fewer weights, for the same output shape.

With the 3 × 3 filters used everywhere else in the network the factor is 29,826,162.

The same output shape, two ways



A logarithmic axis, and the two bars still barely fit on the same figure.

What the saving is *not*

It is not a saving in arithmetic.

- The convolution still computes $C_{\text{out}} HW$ outputs, each a sum over $C_{\text{in}} k^2$ terms. That is the same order of multiplications as the dense layer's, to within the window size
- What shrank is the number of **distinct numbers that have to be stored and learned**

So the win is in memory and in statistics, not in floating-point operations. Confusing the two is how people end up expecting convolutions to be fast.

Weight sharing is a constraint, not a trick

A dense layer *can* represent any convolution: set its weight matrix to the block-Toeplitz matrix that repeats the kernel. Nothing is lost in expressive power.

> **WHAT IS GAINED**
>
> The convolution is that matrix with 5,478,275 tied parameters. Tying them says *a pattern means the same thing wherever it appears* — and that statement is true of photographs and false of, say, a spreadsheet.

This is the bias–variance trade of Lecture 5, made architecturally. We bought a large reduction in variance with an assumption we are prepared to defend.

2 · What that assumption is, precisely

Let $T_{\mathbf{v}}$ be the operator that shifts an image by the vector $\mathbf{v}$:

$$(T_{\mathbf{v}} x)(\mathbf{p}) \;=\; x(\mathbf{p} - \mathbf{v})$$

A map f is **equivariant** to translation when

$$f(T_{\mathbf{v}}\, x) \;=\; T_{\mathbf{v}}\, f(x) \qquad \text{for every } \mathbf{v}$$

Shift then compute, or compute then shift — the same answer.

Convolution is equivariant — two lines

Write the convolution as $(w * x)(\mathbf{p}) = \sum_{\mathbf{u}} w(\mathbf{u})\, x(\mathbf{p} + \mathbf{u})$. Then

$$
\begin{aligned}
(w * T_{\mathbf{v}} x)(\mathbf{p}) &= \sum_{\mathbf{u}} w(\mathbf{u})\, x(\mathbf{p} + \mathbf{u} - \mathbf{v}) \\
&= (w * x)(\mathbf{p} - \mathbf{v}) \;=\; \big(T_{\mathbf{v}}(w * x)\big)(\mathbf{p})
\end{aligned}
$$

One change of variable. The proof works because w does not depend on $\mathbf{p}$ — **which is exactly what weight sharing is**. Untie the weights and the middle equality fails.

Where the equality is not exact

- **At the border.** Zero padding invents input that was not there, so a shift moves real content into invented content. The identity holds on the interior and nowhere else
- **Under stride greater than one.** The output grid samples every s-th position, so equivariance survives only for shifts that are multiples of s
- **In floating point**, if the two computations happen to sum the same terms in different orders

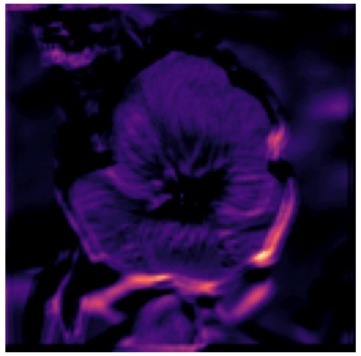
All three are why the next slide states a measurement rather than an assertion.

Shift the flower; the map follows

The map did not change — it moved, by exactly the same 16 pixels

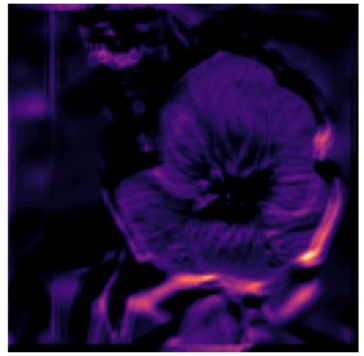
input

feature map, filter 19



input shifted 16 px

its feature map



The map did not change shape. It moved, by exactly the 16 pixels the input moved.

Measured, on the layer you trained

```
y            = conv(x)
y_from_shift = conv(torch.roll(x, 16, dims=3))
y_then_shift = torch.roll(y, 16, dims=3)

m = 24                                        # drop the border and the seam
d = (y_from_shift - y_then_shift)[..., m:-m, m:-m].abs().max()
```

	Value
largest absolute difference, interior	**0** ✓
largest activation, same region	4.15

Bit-for-bit identical, not merely close. On this backend the two computations perform the same additions in the same order, so even the floating-point caveat does not bite — but check it rather than assuming it, because on another backend it will not be exactly zero.

3 · Equivariance is not invariance

A map g is **invariant** to translation when

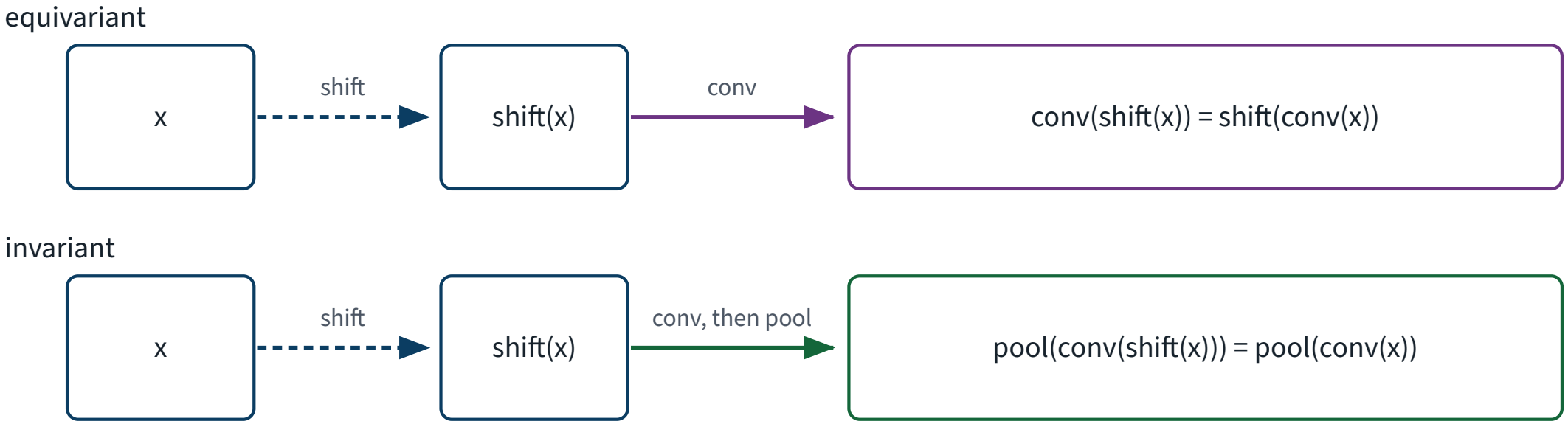
$$g(T_{\mathbf{v}}\, x) \;=\; g(x) \qquad \text{for every } \mathbf{v}$$

	Under a shift the output…	Keeps	?
✓ equivariant	**✓ moves with the input**	**✓ yes**	
invariant	does not change	**✗ no**	

Invariance is strictly the stronger and strictly the lossier of the two. It is the one you get by *discarding* the equivariant structure.

The two laws, side by side

Equivariance moves the answer · invariance forgets where it was



Classification wants the second line. Per-pixel prediction cannot afford it.

Pooling is what turns the first line into the second.

How pooling manufactures invariance

A 2×2 max pool is invariant to any shift that keeps the maximum inside its own window — so half the one-pixel shifts, in each direction.

- Stack four pooling stages and the tolerated shift grows to $\pm$ 8 pixels in the last feature map
- A global pool over the whole map is invariant to *every* shift — there is no position left to be sensitive to

The invariance is approximate and it is built, one factor of two at a time. Nothing in the architecture declares it.

Measured, on the same shift

Cosine similarity between the representation of the image and the representation of the image shifted 16 pixels:

Representation	Cosine similarity
✗ the spatial feature maps, flattened	✗ 0.6379
✓ after a global max pool	1.0000 ✓

The map changed by 36.2%. The pooled vector changed by 0.0004%.

Why classification wants the second line

“This is a sunflower” is true wherever the sunflower is.

- The label is a function of the content, not of the position, so a representation that still carries the position is carrying a nuisance variable
- Discarding it is a reduction in variance, paid for with no bias — provided the assumption really holds

The dense head after your four pooling stages sees a 8 × 8 grid rather than a single vector, so the invariance is partial. That is a design choice, and a defensible one.

Why per-pixel prediction cannot afford it

Segmentation asks a different question: *for every pixel, which class is it?* The answer **is** the position.

	Our network
Input grid	128 × 128
Last feature map	8 × 8
✗ One cell answers for	**✗ 256 pixels**
✗ Spatial resolution lost	**✗ 256×**

Two flower boundaries 15 pixels apart are the same cell. Lecture 14 has to put that resolution back, and the whole of its architecture is about how.

Where the thread has got to

- Weight sharing is a constraint that costs 5,478,275-fold fewer parameters
- It is exactly the condition under which convolution commutes with translation
- Pooling converts that commutation into indifference, which classification wants and per-pixel prediction cannot have

One thing left, and it is the one that stops your Colab session.

4 · A question with a wrong answer everyone gives

Your network has 4,807,494 parameters and your session died with `out of memory`.

Which of those two facts caused the other?

The parameters, in bytes

What	Bytes each	Total
weights	4	18.3 MB
gradients, one per weight	4	18.3 MB
Adam's first moment	4	18.3 MB
Adam's second moment	4	18.3 MB
✓ everything parameter-shaped	16	**73.4 MB ✓**

Sixteen bytes per parameter, not four. That factor is the first thing people forget, and it is still not the answer.

What else is alive during a backward pass

Lecture 10's derivation, in bytes: the reverse sweep evaluates each Jacobian *at the value the forward pass reached*.

$$\bar{u}_{\text{in}} = \bar{u}_{\text{out}} \left. \frac{\partial f}{\partial u_{\text{in}}} \right|_{u_{\text{in}}}$$

✗ So every layer's output stays in memory from the moment it is computed until the backward pass reaches it.

That is not an implementation detail of PyTorch. It is what reverse-mode automatic differentiation is.

Two boxes, one of them multiplied by the batch

What is alive between the forward and the backward pass

parameters

weights + gradients + Adam

fixed — independent of the batch

every layer's output

kept because the backward pass evaluates each Jacobian at the value the forward pass reached — Lecture 10, in bytes

× batch size

→

out of memory

halve the batch, not the model

Only one of these two boxes has the batch size in it.

Count them, for the actual network

```
x, total = torch.zeros(1, 3, 128, 128), 0
for m in net:
    x = m(x)
    total += x.numel()

print(f"{total:,} floats per image")
print(f"{total * 4 / 2**20:,.1f} MB per image")
```

	Per image
stored activations	5,964,646
✗ at four bytes each	✗ 22.8 MB

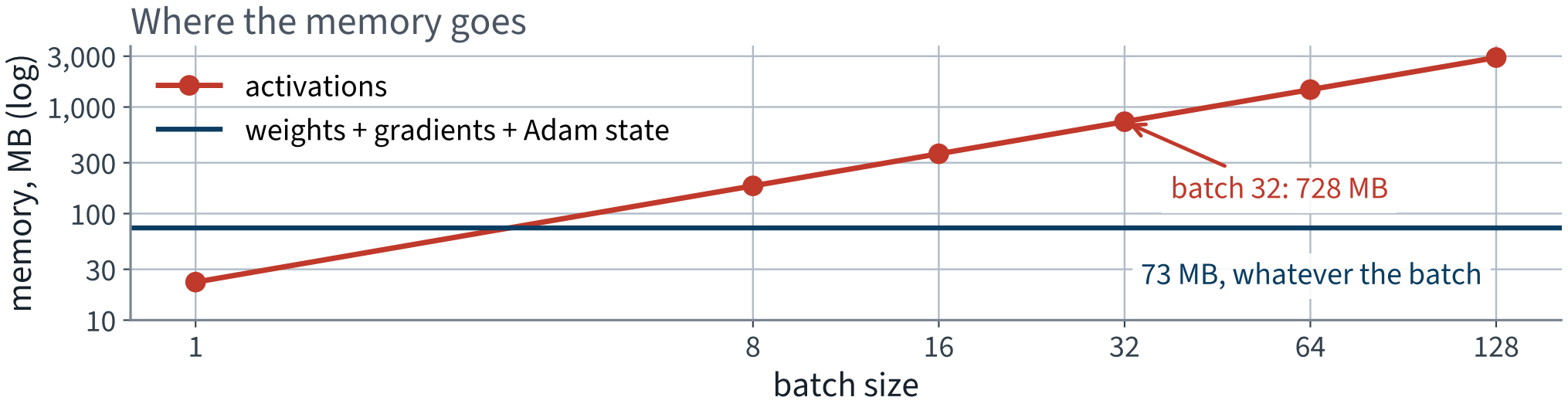
The split, at the batch size you used

	MB at batch 32
weights	18.3
weights + gradients + Adam	73.4
✗ activations	**✗ 728**

9.9×

the activations are that many times everything parameter-shaped, put together — and about forty times the weights alone.

One of them is a flat line



The parameter line does not move. The activation line is a straight line through the origin, and it wins almost immediately.

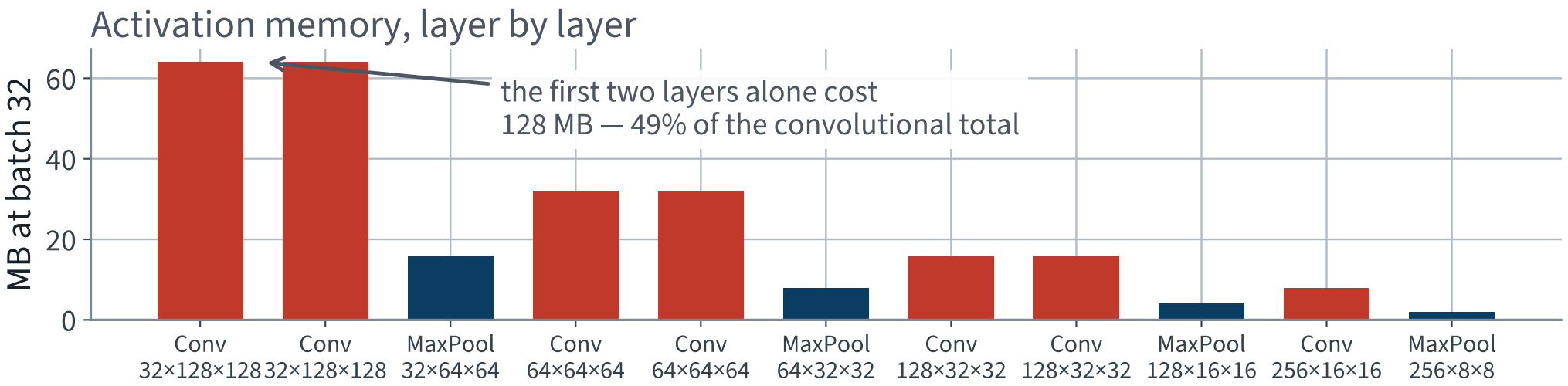
And it is not spread evenly

The activation cost of a layer is $C \times H \times W$. Channels double as the map quarters, so **the total halves at every pooling stage**.

Layer	Output	MB at batch 32
✗ first convolution	32 × 128 × 128	**✗ 64**
third convolution	64 × 64 × 64	32
last convolution	256 × 16 × 16	8

The layers nearest the image are the expensive ones, and they are the ones with almost no parameters.

The bill, layer by layer



49% of the convolutional activation memory is in the first two layers.

Parameters and activations live at opposite ends

	Concentrated in	Because
✗ activations	✗ the *early* layers	H and W are still large
✓ parameters	✓ the *late* layers	$C_{\text{in}}C_{\text{out}}$ is large, and k is fixed

Two different bills, sent by two different halves of the same network.

This is why “make the model smaller” and “make the run fit in memory” are different problems with different answers. It comes back in ten minutes, when we choose which block of a pretrained network to unfreeze.

Check the arithmetic against the allocator

```
torch.mps.empty_cache()
base = torch.mps.current_allocated_memory()   # model + optimiser state
out  = net(x)                                 # batch of 32, training mode
torch.mps.synchronize()
print((torch.mps.current_allocated_memory() - base) / 2**20, "MB")
```

	MB
predicted, by counting outputs	728
✓ measured, by the backend	**568 ✓**

The measurement is lower than the prediction: the allocator frees an activation as soon as nothing can still need it, and in a plain `Sequential` the ReLU outputs of already-consumed layers qualify. The prediction is the ceiling, which is the number you want when sizing a run.

The consequence, as a rule

> **WHEN A RUN RUNS OUT OF MEMORY**
>
> **Halve the batch, not the model.** Activation memory is linear in the batch size and the parameter memory does not move at all, so the first lever is the one that costs you nothing but wall clock.

In order: smaller batch, then smaller input resolution (quadratic), then gradient checkpointing, then mixed precision. Reducing the parameter count is near the bottom of the list — and on this network it would mostly delete the dense head, which is not where the memory is.

What the derivation buys you, in four lines

1. Weight sharing costs 5,478,275-fold fewer parameters than the dense layer with the same output shape
2. It is precisely the condition for translation equivariance, and the proof is a change of variable
3. Pooling trades equivariance for invariance — wanted here, fatal in two lectures' time
4. Parameters are 18.3 MB; activations at batch 32 are 728 MB

Point 1 is also, as it turns out, the reason the repair works.

THE METHOD

Convolution, pooling, and an architecture

40 minutes · examinable

The idea, in one sentence

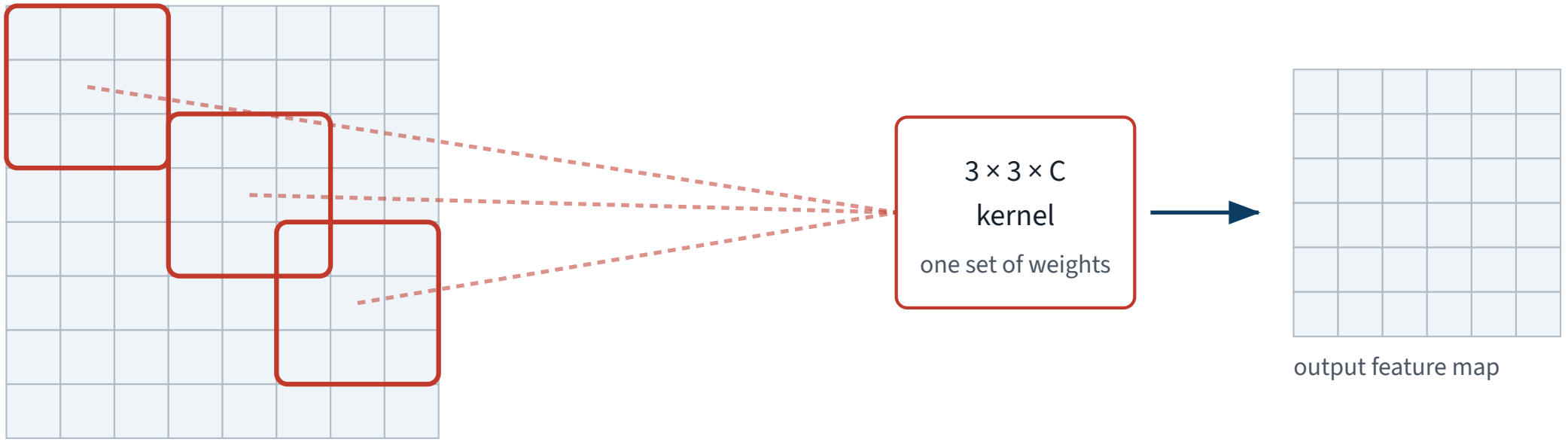
A feature worth detecting in the top left corner is worth detecting in the bottom right, so **use the same weights everywhere**.

- That is the whole of it. Everything else is bookkeeping
- It buys two things at once: far fewer parameters, and a network that does not have to relearn a petal edge at every position

Both halves were proved in the mathematics block. Here we use them.

One filter, every position

One filter, every position, the same weights



A dense layer would learn a different weight for every one of those windows.

Three windows, one set of weights, one output map. A dense layer would learn a separate weight for every window.

A convolution, as arithmetic

For an input x with C channels and a kernel w of size $k \times k$, the output at position (i, j) of filter f is

$$z_{ij}^{(f)} = b^{(f)} + \sum_{c=1}^{C} \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} w_{c,u,v}^{(f)} \, x_{c,\, i+u,\, j+v}$$

A dot product between the kernel and a window of the input, repeated at every position. The sum over c is why a filter reads *all* the input channels at once.

The three numbers that describe a convolution

Name	What it sets	Ours
kernel size k	how much of the input one output sees	7, then 3
stride s	how far the window moves between outputs	1
padding p	how many zeros ring the input	$k/2$, rounded down

$$H_{\text{out}} = \left\lfloor \frac{H_{\text{in}} + 2p - k}{s} \right\rfloor + 1$$

With $s = 1$ and $p = \lfloor k/2 \rfloor$ the output has the same height and width as the input. That is the setting we use everywhere.

`nn.Conv2d`, and the defaults you did not ask for

```
nn.Conv2d(
    in_channels=3,        # must match the previous layer's output
    out_channels=32,      # how many filters -> how many output maps
    kernel_size=7,
    stride=1,             # the default
    padding=3,            # NOT the default; the default is 0
    bias=False,           # a batch-norm follows, and it has its own shift
)
```

Reviewer question 5, every time: **padding defaults to zero**, which silently shrinks every feature map by $k - 1$ and will eventually give you a shape error twelve layers later.

Filters in, feature maps out

- One filter sweeps the whole image and produces **one** two-dimensional map: how strongly that pattern fired, everywhere
- 32 filters produce 32 maps, stacked into a $32 \times H \times W$ tensor
- The next layer's filters read all 32 of them at once

$$\underbrace{3}_{\text{channels in}} \rightarrow \underbrace{32}_{\text{filters}} \rightarrow \underbrace{32}_{\text{channels out}}$$

“Channel” and “feature map” are the same object seen from the two sides of a layer.

How many parameters is that

$$\#\text{weights} = C_{\text{out}} \times C_{\text{in}} \times k \times k$$

Our first layer: $32 \times 3 \times 7 \times 7 = 4,704$ weights.

Notice what is *not* in that formula.

✓ The image size. A convolutional layer has the same number of parameters on a 128-pixel image as on a four-megapixel one.

Pooling

After a couple of convolutions, halve the height and width by keeping only the largest value in each 2×2 block.

- Three quarters of the activations disappear, and with them three quarters of the memory and the arithmetic in every later layer
- It has **no parameters** — nothing about it is learned
- Each later unit therefore sees a larger piece of the original image

There is a fourth consequence — invariance — and it is the one the mathematics block spent twenty minutes on.

Max pooling, on numbers

4 × 4 feature map

5	3	8	1
2	9	4	0
1	6	7	3
4	2	5	8

after 2 × 2 max pool

9	8
6	8

Sixteen numbers in, four out, and the four are the largest of each block.

Pooling in code

```
>>> a = torch.tensor([[[[5., 3., 8., 1.],
...                    [2., 9., 4., 0.],
...                    [1., 6., 7., 3.],
...                    [4., 2., 5., 8.]]]])
>>> nn.MaxPool2d(2)(a)
tensor([[[[9., 8.],
          [6., 8.]]]])
```

`kernel_size=2` sets the stride to 2 as well, so the windows do not overlap. That is the default, and it is the one you want.

Why every convolution is followed by a batch norm

Because you proved in the previous lecture that it has to be.

- Ten layers of untreated convolutions on 1,020 images do not train: the loss sits at $\log 102 \approx 4.62$ and the network predicts one class
- `BatchNorm2d` normalises across the batch *and* both spatial axes, so it holds one mean and one variance per channel

```
nn.Conv2d(c_in, c_out, 3, padding=1, bias=False)
nn.BatchNorm2d(c_out)     # 2 * c_out parameters, not 2 * c_out * H * W
nn.ReLU()
```

The architecture

Stage	Layers	Output
1	conv 7×7 ×32, conv 3×3 ×32, pool	32 × 64 × 64
2	conv ×64, conv ×64, pool	64 × 32 × 32
3	conv ×128, conv ×128, pool	128 × 16 × 16
4	conv ×256, pool	256 × 8 × 8
✓ head	**✓ flatten, dense 256, dropout, dense 102**	**✓ 102**

Channels double as the map halves — the textbook's own rule of thumb, and the reason the memory does not collapse to nothing.

The architecture, in code

```
def conv_block(c_in, c_out, k=3):
    return [nn.Conv2d(c_in, c_out, k, padding=k // 2, bias=False),
            nn.BatchNorm2d(c_out), nn.ReLU()]

net = nn.Sequential(
    *conv_block(3, 32, k=7), *conv_block(32, 32), nn.MaxPool2d(2),
    *conv_block(32, 64),     *conv_block(64, 64), nn.MaxPool2d(2),
    *conv_block(64, 128),  *conv_block(128, 128), nn.MaxPool2d(2),
    *conv_block(128, 256),                        nn.MaxPool2d(2),
    nn.Flatten(),
    nn.Linear(256 * 8 * 8, 256), nn.ReLU(),
    nn.Dropout(0.5),
    nn.Linear(256, 102),
)
```

Thirty modules. Nothing in it that was not on a slide in this lecture or the previous one.

Check the shapes before you check anything else

```
x = torch.zeros(2, 3, 128, 128)
for m in net:
    x = m(x)
    print(f"{type(m).__name__:12s} {tuple(x.shape)}")

assert x.shape == (2, 102), f"the head is wrong: {x.shape}"
```

Reviewer question 3, and it costs four lines. A shape error inside a training loop surfaces as a size-mismatch traceback two hundred lines deep; here it surfaces as a printed line.

Count the parameters, and look at where they are

Part	Parameters	Share
Every convolution and batch-norm layer together	586,720	12.2%
✗ One dense layer, 16,384 → 256	**✗ 4,194,560**	**✗ 87.3%**
Output layer, 256 → 102	26,214	0.5%
✓ Total	**✓ 4,807,494**	100%

✗ Nine parameters in ten are in the layer that is not convolutional.

Why that number should bother you

4,807,494 parameters. 1,020 training images.

4713

parameters per training image.

There is nothing in the architecture stopping this network from storing the training set. Dropout at 0.5 is the only thing we have asked to prevent it, and it will not be enough.

The training loop — unchanged

```
opt   = torch.optim.Adam(net.parameters(), lr=3e-4)
lossf = nn.CrossEntropyLoss()

for epoch in range(80):
    net.train()
    for xb, yb in train_loader:
        opt.zero_grad()
        loss = lossf(net(xb), yb)
        loss.backward()
        opt.step()
```

Exactly the five lines from Lecture 10. Convolution changed the model; it did not change the loop, because autograd does not care what the modules are.

One hyperparameter that is not a free choice

Learning rate	Mean test accuracy	sd over seeds
✗ 10^{-3}, Adam's default	✗ 2.7%	1.31 pts
✓ 3×10^{-4}	14.7% ✓	2.59 pts

Measured on this architecture and this data, at 40 epochs. The default does not diverge — it plateaus low, which is the failure mode that does not announce itself.

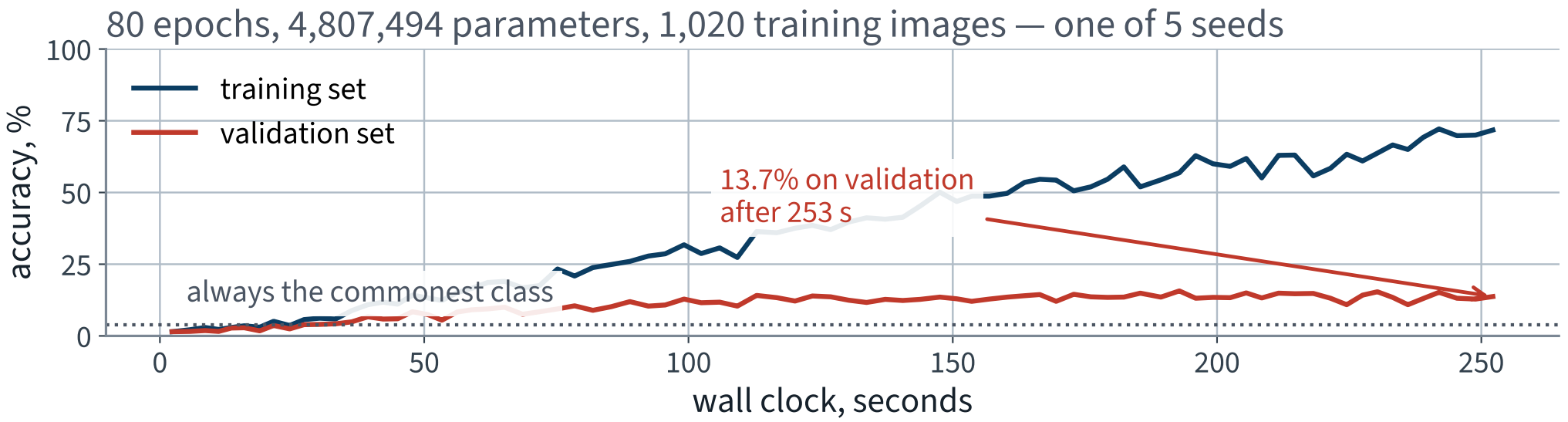
12.1 points, from one keyword argument nobody wrote down.

Run it

Setting	Value
Epochs	80
Batch size	32
Steps per epoch	32
Total optimiser steps	2,560
✗ Wall clock, seed 42	**✗ 253 s**

Measured on an Apple MPS backend, no CUDA, and including a validation pass after every epoch. A free Colab T4 is in the same range for a model this size; the point of the number is the comparison we make next lecture, not the hardware.

The cost, on the x-axis



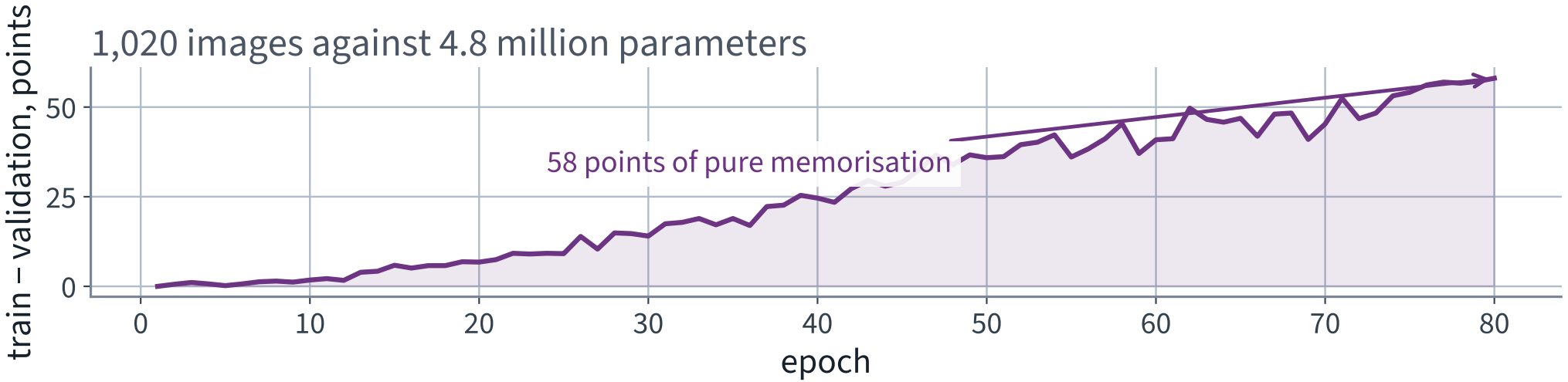
The x-axis is seconds, not epochs. That is deliberate: from the next lecture on, time is one of the things being compared.

Reading it

- The training curve reaches 71.8% — the network is well on its way to memorising 1,020 images
- The validation curve stops climbing early and then does very little for the rest of the run
- The last two thirds of the wall clock bought almost nothing

Both curves are correct. They are describing two different things.

The gap, on its own axis



58 points of the training accuracy is memorisation. Lecture 5 named this shape: a persistent gap is variance.

The test set. Once.

```
net.eval()
with torch.no_grad():
    correct = sum((net(xb).argmax(1) == yb).sum().item()
                  for xb, yb in test_loader)
print(f"test accuracy {correct / len(test_ds):.4f}")
```

15.3%

Mean over 5 seeds, standard deviation 2.34 points, range 11.0% to 18.1%. One run of a network this size on 1,020 images is an anecdote.

Why five runs and not one

Seed	Test accuracy
42	11.0%
43	15.7%
44	18.1%
45	16.1%
46	15.7%

✗ 7.1 points between the best and the worst, and nothing changed but the seed.

Any comparison you make from here that is smaller than that is not a comparison.

Where that number sits

Model	Test accuracy	Wall clock
Uniform random guess	0.98%	—
Always the commonest species	3.87%	—
✓ Convolutional network, from scratch	**15.3% ✓**	**✗ 364 s**
What the operator needs	90%	—

4.0 times the majority baseline — and a long way from deployable.

The 364 s is the mean over five seeds, including the validation passes. The 253 s two slides ago is the seed-42 run on its own. Lecture 13 compares against this column, so it is the one to carry forward.

Look at what it learned

The first layer's weights are 4,704 numbers arranged as 32 filters of $7 \times 7 \times 3$. That is small enough to *look at*.

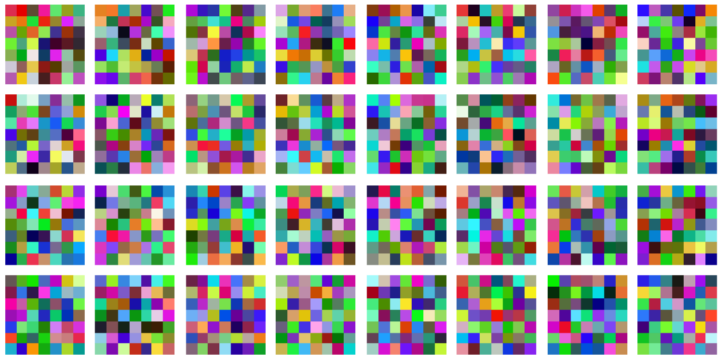
```
w = net[0].weight.detach().cpu()      # (32, 3, 7, 7)
lo = w.amin(dim=(1, 2, 3), keepdim=True)
hi = w.amax(dim=(1, 2, 3), keepdim=True)
grid = ((w - lo) / (hi - lo)).permute(0, 2, 3, 1)   # ready for imshow
```

Each filter is rescaled to its own range, because otherwise the strongest filter is the only one you can see. Keep the initialisation, so there is something to compare against.

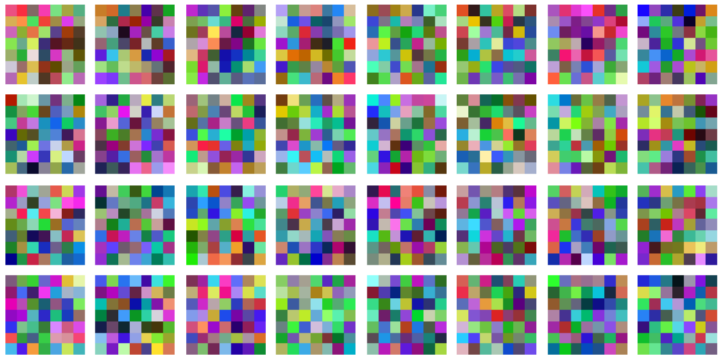
Thirty-two filters, before and after

All 32 filters of the first layer, 7 × 7 × 3, each rescaled to its own range

at initialisation



after 80 epochs



Look at the two blocks for ten seconds before the next slide, and say what changed.

It moved a little. It did not become anything

```
import torch.nn.functional as F

before, after = init_filters.flatten(), net[0].weight.flatten()
print(f"cosine similarity {F.cosine_similarity(before, after, dim=0):.4f}")
print(f"mean |change| / mean |weight| "
      f"{(after - before).abs().mean() / before.abs().mean():.1%}")
```

	Value
✓ cosine similarity, all 4,704 weights before against after	**0.9694 ✓**
mean absolute change, as a share of the mean weight	23.9%
weights that changed sign	247

✗ The weights moved. No structure appeared — the right-hand block is still noise, and 253 seconds of training bought it.

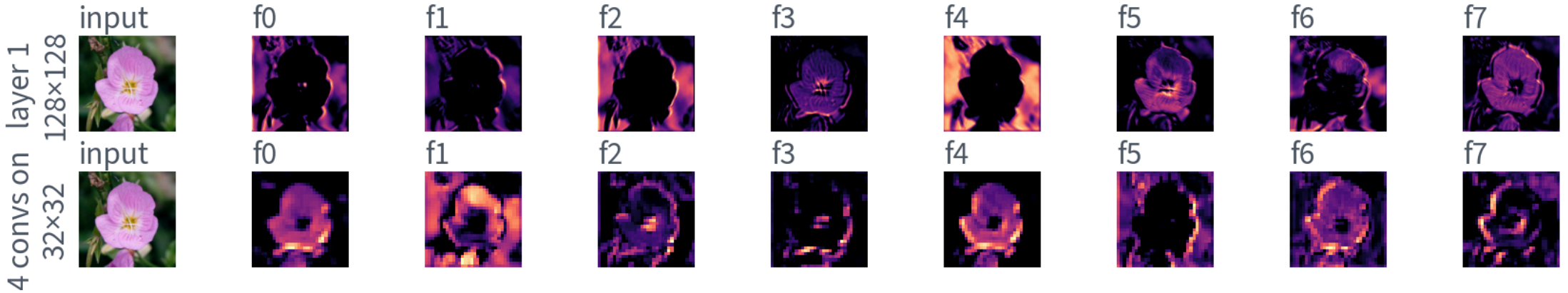
Two reasons, and both are worth knowing

1. **The gradient that reaches it is small.** Ten layers of chain rule separate the loss from these weights, and 1,020 images give 2,560 optimiser steps in total
2. **Its scale does not matter.** A batch norm follows immediately, and normalising the output makes the overall size of these weights irrelevant to the function — so there is no gradient pressure on it at all

The textbook picture of a first layer — colour blobs, oriented edges — is real, and you will see it next lecture. It comes from a corpus a thousand times this size.

What a filter does to one photograph

Feature maps: the first layer keeps the picture, the fourth keeps a summary



Early maps still look like the photograph. Four layers down they do not, and that is the point.

Reading a feature map

- Bright means *this filter fired strongly here* — nothing more mystical than that
- The spatial layout survives: a bright patch is where in the image the pattern was found
- After four pooling stages the map is 8×8, so one cell answers for a 256-pixel square of the original

That last line is a loss of information we chose. Next lecture asks what we lost and whether we wanted to.

THE LAST FIFTEEN MINUTES

What it costs, and what it cannot do

15 minutes

The scoreboard

	Test accuracy
Uniform guess	0.98%
Commonest species	3.87%
✓ Yours, today	**15.3% ✓**
Operator's requirement	90%

Write your measured number on the same sheet, next to the one you predicted.

What we deliberately did not do

- No data augmentation — the network saw exactly 1,020 images, 80 times each
- No pretrained weights — every one of the 4,807,494 parameters started at random
- No learning-rate schedule beyond a constant 3×10^{-4}
- No architecture search — one architecture, chosen and kept

Each of those is a lever. Two of them move the number a great deal, and one of the two moves it by more than everything else in this course put together.

What this number is and is not

- **Is**: the mean of 5 runs of one architecture on one split, each measured on 6,149 held-out images
- **Is not**: a claim about convolutional networks. It is a claim about this one, trained on 1,020 images
- The seed-to-seed standard deviation was 2.34 points. Any comparison smaller than that is not a comparison

Quote the spread whenever you quote the number. It is the difference between a measurement and an anecdote.

The vocabulary, in one place

Term	Means
filter / kernel	the $C \times k \times k$ block of weights that is shared across positions
feature map	the two-dimensional output of one filter over the whole input
channel	a feature map, seen from the next layer
stride	how far the window moves between outputs
padding	zeros added around the input so the output keeps its size
receptive field	how much of the original image one unit can see

Receptive field, since it is on the list

One unit of the first layer sees 7×7 pixels. Stack a 3×3 convolution on it and the unit sees 9×9; pool, and the next one sees twice as much again in each direction.

$$r_\ell = r_{\ell-1} + (k_\ell - 1) \prod_{i<\ell} s_i$$

By the last convolution a unit sees most of the image. That is why a network with 3×3 filters can recognise an object larger than any single filter — depth buys extent.

Five ways this build goes wrong quietly

Mistake	Symptom
`padding=0` left at the default	maps shrink; a shape error many layers later
Forgetting `net.eval()`	the score wobbles between calls — Lecture 10
Normalising with test statistics	measured today at 0.25 points
`lr` left at Adam's default	12.1 points, and no error
Pooling before any convolution	throws away the pixels before anything reads them

Why not simply make it deeper

Because you tried that in Lecture 11, and because there is nothing left to learn from.

- The training accuracy on the plotted run is 71.8%. More capacity fits the same 1,020 images harder
- The gap is 58 points. That is a variance problem, and Lecture 5 says variance is fixed with more data or more constraint, not more parameters

Which of those two we can actually get more of is the question the next lecture answers.

Scope

Topic	Chapter
Convolutional layers, filters, feature maps	12
Stacking multiple feature maps	12
Pooling layers	12
A CNN architecture for image classification	12
Batch normalisation, Adam, dropout	11
Colab runtimes, dataset caching	not examinable

The notebook for this lecture

Open Lecture 12 in Colab

- **Runs on free CPU.** Flowers102 is subsampled and the network is small; a few minutes end to end
- The dense-against-convolutional parameter count, computed rather than quoted — the arithmetic you can do on paper first
- Equivariance checked numerically: shift the input, shift the feature map, and compare

> **WHAT TO CHANGE**
>
> Replace the first convolution with a dense layer of the same output width and print the parameter count. The ratio is the derivation, measured.

What we did

1. Counted the parameters of a dense layer and a convolutional layer on the same image, and found the ratio does not depend on the image size
2. Proved convolution is translation *equivariant*, and saw where that fails — at the boundary, and under anything that is not a translation
3. Separated equivariance from invariance, and saw that pooling buys invariance: what classification wants and segmentation cannot afford
4. Counted the memory and found that **activations, not parameters**, are what exhaust a device — and that the two scale differently
5. Built a convolutional network, trained it, and read its filters and its feature maps
6. Measured what it cost, and named what it still cannot do

NOT PRESENTED — FOR AFTERWARDS

Exercises Lecture 12

five, in the style of the written paper

Exercises — Lecture 12

1 A dense layer mapping a $3 \times 128 \times 128$ input to a $32 \times 128 \times 128$ output has $n_{\text{in}} n_{\text{out}}$ weights. Give the convolutional count for a 7×7 kernel, and say which symbols are absent from it. [5]

2 State the difference between equivariance and invariance, and say which of the two segmentation cannot afford. [4]

3 Convolution is equivariant to translation on the interior of an image but not at the border. Give the reason. [3]

Solutions on Lecture 13’s deck.

Exercises — Lecture 12 (continued)

4 A training run fails with out-of-memory. State which of parameters and activations usually dominates, and give the first thing to change. [4]

5 Parameters and activations are anti-correlated across a convolutional stack. State where each is concentrated, and what that means for any intuition carried over from dense networks. [4]

Solutions on Lecture 13’s deck.

THE FIVE SET AT THE END OF LECTURE 11

Solutions Lecture 11

not part of this lecture

Solutions — Lecture 11

1. One layer multiplies the standard deviation of the backward signal by…

✓ They multiply it by ρ^{L-1} — $L-1$ steps between the first gradient and the last — so anything but 1 vanishes or explodes geometrically.

- A constant applied L times is exponential in L
- There is no regime where a repeated constant other than 1 is safe

2. Preserving the forward signal asks for $\mathrm{Var}(w) = 1/n_{\text{in}}$ and preserving the gradient…

✓ They agree only when $n_{\text{in}} = n_{\text{out}}$; Glorot takes the harmonic mean, $2/(n_{\text{in}} + n_{\text{out}})$.

- Any layer that changes width cannot satisfy both
- The compromise accepts a small error in each direction rather than a large one in either

Solutions — Lecture 11 (2)

3. He initialisation doubles the weight variance for ReLU. Derive the factor of two in one line.

✓ ReLU zeroes half of a symmetric zero-mean input, so $\mathbb{E}[\mathrm{ReLU}(z)^2] = \frac{1}{2}\mathbb{E}[z^2]$.

- Exactly half the variance is discarded at each layer
- Doubling $\mathrm{Var}(w)$ puts it back

4. A gradient profile that is a straight line on a log axis tells you something a curved one does not. State…

✓ The attenuation is the same factor at every layer, so it is the initialisation rather than one bad layer.

- A constant ratio per layer is a straight line in log space
- A single broken layer would show a step, not a slope

Solutions — Lecture 11 (3)

5. Gradient clipping is applied to a network whose gradients are fifteen orders of magnitude too small. Predict…

✓ No effect — clipping bounds gradients that are too large.

- The threshold is never reached, so nothing is rescaled
- Reaching for it means the failure mode was never measured

LECTURE 13 · GÉRON CH. 12

Transfer learning

Why features learned on one corpus transfer, and how far down the stack that holds
Freezing, progressive unfreezing, differential learning rates
The same accuracy as today, in a fraction of the training

Run the Lecture 12 notebook first